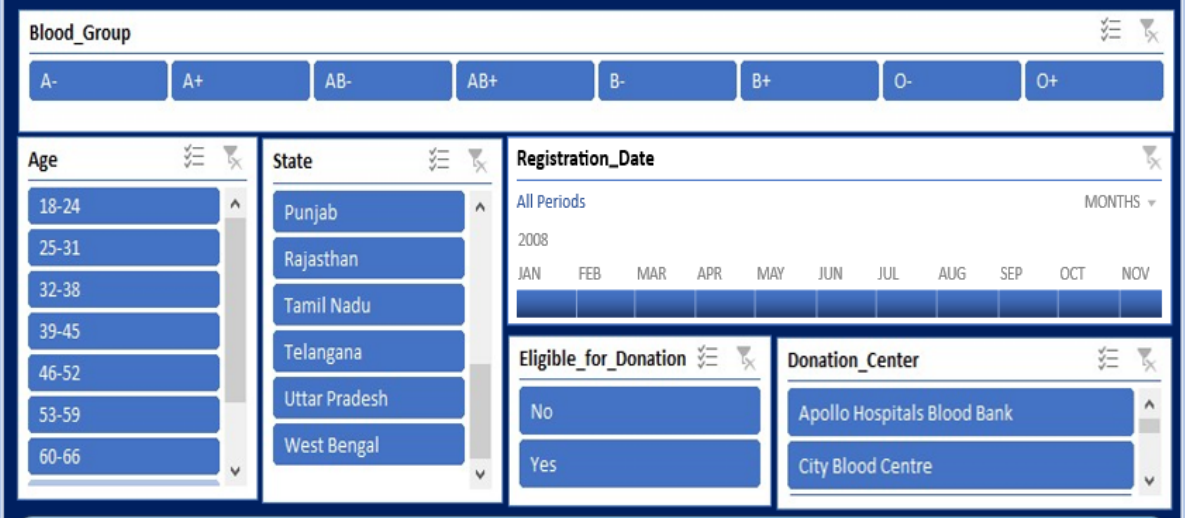
Blood Donation Analytics

Total Donors	Top Donors	Most Donated	Top City	Avg. Donations
10000	Female	32-38	Maharashtra	2.0848



About

Dashboard

Demographics

Geographics

Blood Donation Analytics Dashboard

Project Overview

This dashboard provides a comprehensive, data-driven overview of a regional blood donation program. By analyzing a dataset of **9,000+ registered donors** across multiple Indian states, this tool transforms raw administrative data into actionable strategic intelligence. It

1. The "Silent Inventory" Risk Problem

Problem: Blood banks often operate reactively, only discovering a shortage of a specific blood type (e.g., AB-) during an emergency. Without real-time data, they cannot predict supply gaps for rare blood groups.

Dashboard Solution: By visualizing donor counts by blood type and filtering by eligibility, the dashboard immediately flags which blood groups are critically low and require urgent recruitment.

2. Blood Type Supply Risk

The Problem:

While O+ is the most common and most demanded blood type, rare types like **B-, AB-, and AB+** have a very small donor pool, making every single donor critical. Without real-time visibility, blood banks only realize a rare type shortage during an emergency, leading to cancelled surgeries, delayed transfusions, and panicked public appeals.

Dashboard Solution:The dashboard flags rare blood type donors and tracks their individual eligibility status, allowing for "high-touch".

3. Demographic Targeting Blindness

Problem: Generic "everyone donate" campaigns are inefficient. Blood banks lack data on which age/gender groups are most likely to be eligible and willing to donate.

Dashboard Solution: Age and gender breakdowns help identify the most active demographics (e.g., females 25-35), enabling hyper-targeted communication and camp planning.

4. Geographic & Center Underperformance

Problem: Blood banks often distribute resources (mobile vans, marketing spend) evenly across all cities and centers, ignoring wide disparities in donor participation and eligibility rates.

Dashboard Solution: City-wise and center-wise filters reveal which locations have low donor density or high ineligibility rates, allowing management to shift resources to high-potential areas.

Core Insights from Data

Untapped Pool: A significant number of registered donors have *never* donated but are eligible. Immediate campaign opportunity.

The Eligibility Gap: ~20-25% are ineligible, driven by Hypertension/Diabetes (males) and Anemia (females).

Loyalty Crisis: Majority are 1-time donors. Retention is the biggest growth lever.

Regional Disparity: Donor density and eligibility rates vary drastically by city.

Rare Type Alert: Donors for B-, AB-, and AB+ are critically low and require priority stewardship.

Date : 13/04/2026

Project Author:

Chevalla Nithin

- Aspiring Data Analyst

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Age: 18-24 | 25-31 | 32-38 | 39-45 | 46-52 | 53-59 | 60-66

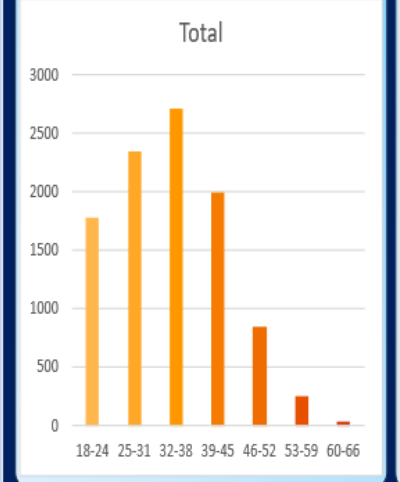
State: Punjab | Rajasthan | Tamil Nadu | Telangana | Uttar Pradesh | West Bengal

Registration_Date: All Periods | MONTHS | 2008 | JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV

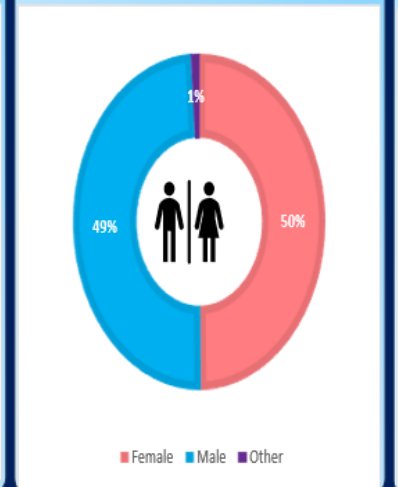
Eligible_for_Donation: No | Yes

Donation_Center: Apollo Hospitals Blood Bank | City Blood Centre

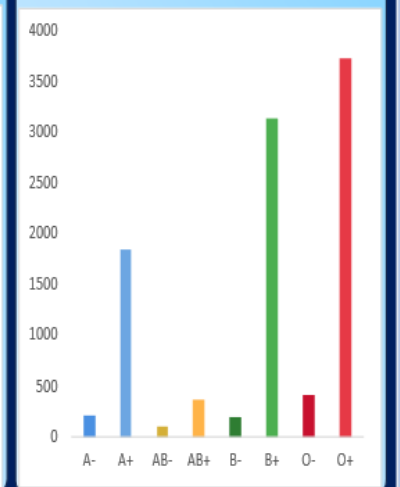
Age Distribution



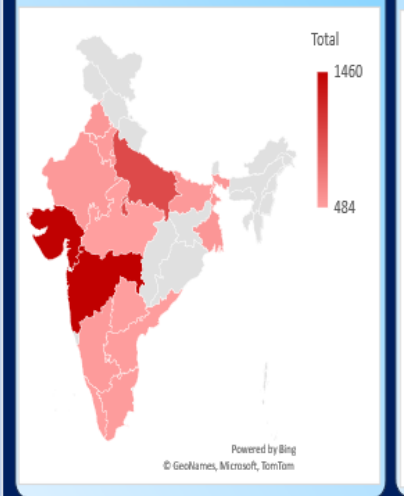
Gender Distribution



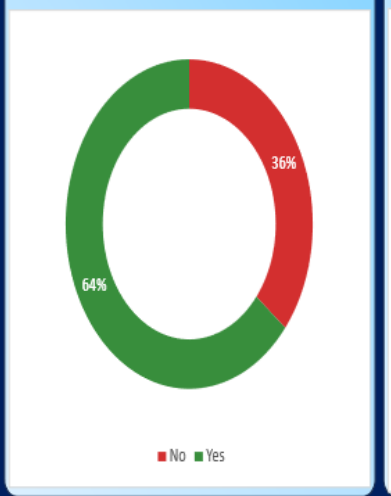
Blood Distribution



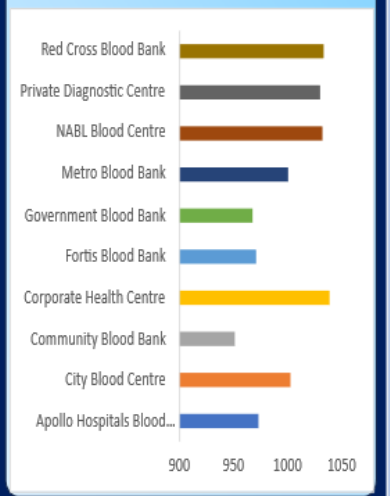
State-Wise Donations



Donar Eligibilty



Donation Centers



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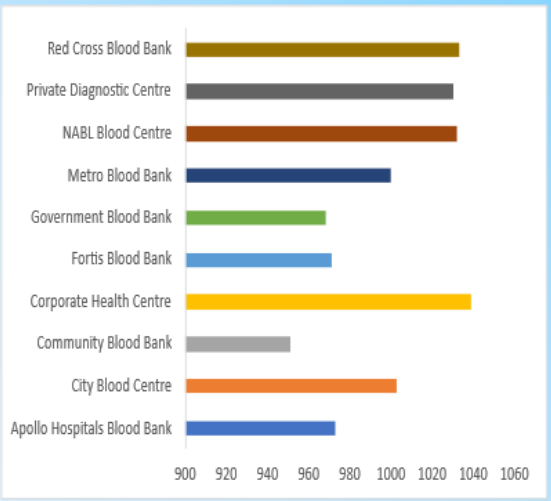

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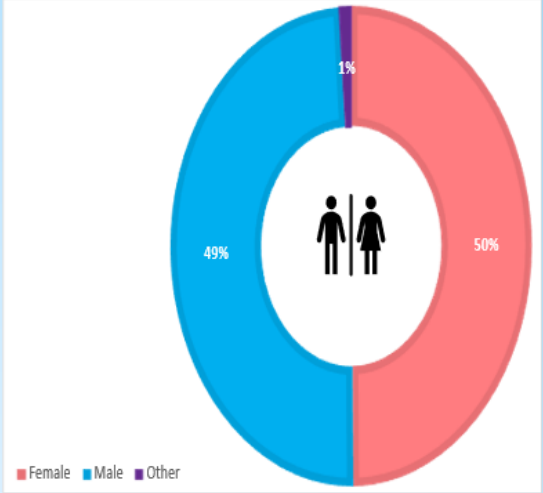
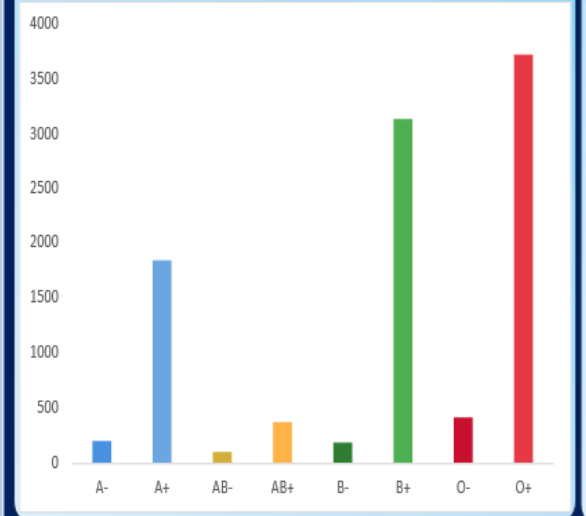


Blood Distribution



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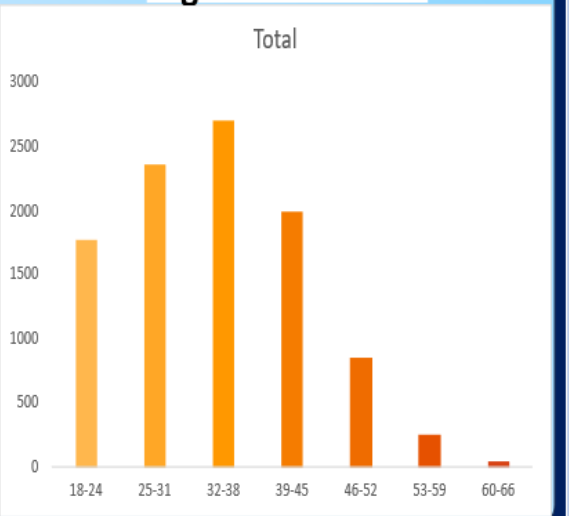

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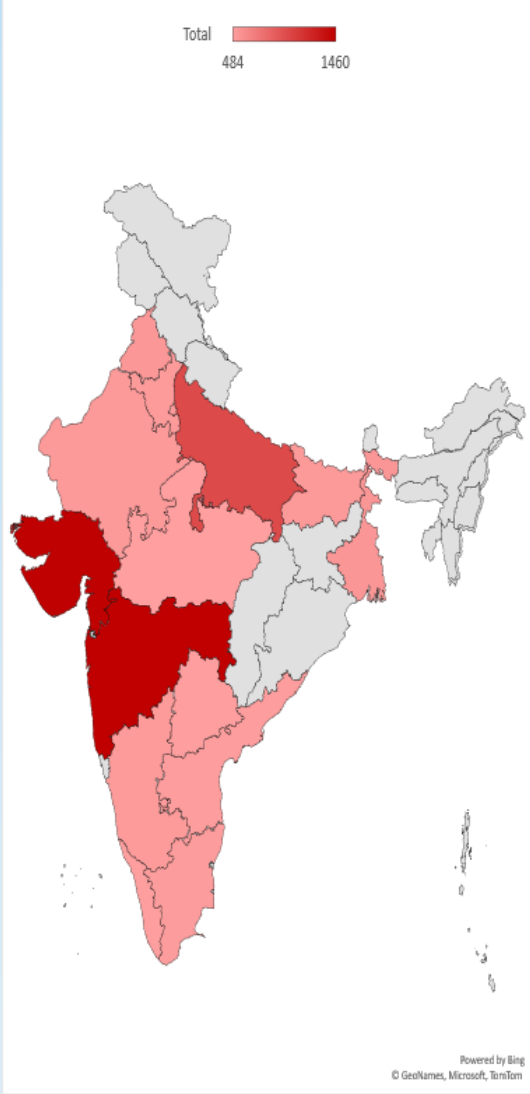
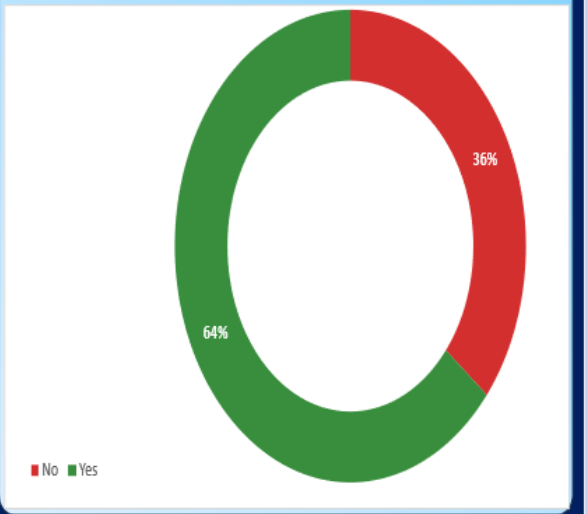
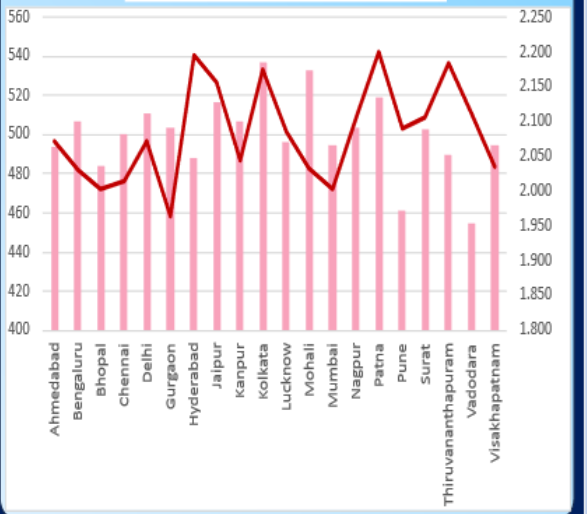


Donar Eligibilty



Average Donations